

COURSE INFORMATION SHEET

Biological Science for Engineers

Course code: **BE24101**

Course title: **Biological Science for Engineers**

Pre-requisite(s): NIL

Co- requisite(s): NIL

Credits: 2 L: 2, T: 0, P: 0

Class schedule per week: 02

Class: B. Tech

Semester: First/Second

Branch: All

Course Objectives

This course enables the students:

1.	To understand fundamental concepts of biology relevant to engineering.
2.	To explore the structure and function of biological molecules and cells.
3.	To learn about genetic principles and molecular biology techniques.
4.	To understand the applications of biological science in various engineering fields considering global challenges and ethical considerations.

Course Outcomes

After the completion of this course, students will be able to:

CO1	Comprehend and apply the fundamental concepts of biological sciences in the context of engineering.
CO2	Analyze the structure and function of biological molecules and cells and their relevance to engineering solutions.
CO3	Demonstrate understanding of genetic principles and molecular biology techniques and their applications in engineering.
CO4	Apply knowledge of biological sciences to innovate and develop solutions in various engineering domains and critically evaluate the role of biological sciences in addressing global challenges, including ethical and safety considerations.

(BE24101) Biological Science for Engineers

Syllabus

Module	Hours
Module I: Introduction to Biological Sciences Overview and importance of biology in engineering, Origin of Life, Cell Theory and Structure	6
Module II: Molecular Biology and Genetics Central Dogma of Molecular Biology, DNA, RNA and Protein structure and function, Mendelian Genetics, rDNA Technology and Genome Editing	6
Module III: Biochemistry Cell Metabolism, Enzymes and Catalysis, Cell Communication and Signalling	6
Module IV: Applications of Biological Sciences in Engineering Biomaterials, Bioinformatics, Biosensors and Bioelectronics (Biological Sensors- Ear & Eye), Synthetic Biology, Nanobiotechnology	6
Module V: Global Challenges and Ethical Considerations Convergence of AI and Biology, Climate change and food security, Biosafety and Biohazards, Ethical Considerations	6

Textbooks

Books Recommended

- **Lehninger A, Principals of Biochemistry**
- **Stryer L, Biochemistry**
- **K. Wilson & K.H. Goulding, A biologist's guide to Principles and Techniques of Practical Biochemistry.**
- **Biology for Engineers" by Arthur T. Johnson**

Reference Books

1. **Purves et al, Life: The Science of Biology**
2. **R. Dulbecco, The Design of Life.**
3. **Biological Science Edited by Soper, Cambridge low price edition.**
4. **Synthetic Biology: A Primer" by Paul S. Freemont and Richard I. Kitney**
5. **"Introduction to Bioinformatics" by Arthur Lesk**
6. **Genomes" by T.A. Brown**

Gaps in the syllabus (to meet Industry/Profession requirements)

POs met through Gaps in the Syllabus

Topics beyond syllabus/Advanced topics/Design

POs met through Topics beyond syllabus/Advanced topics/Design

Course Delivery methods
Lecture by use of boards/LCD projectors/OHP projectors
Tutorials/Assignments
Seminars
Mini projects/Projects
Laboratory experiments/teaching aids
Industrial/guest lectures
Industrial visits/in-plant training
Self- learning such as use of NPTEL materials and internets
Simulation

Course Outcome (CO) Attainment Assessment tools & Evaluation procedure

Direct Assessment

Assessment Tool	% Contribution during CO Assessment
Mid Sem Examination Marks	25
End Sem Examination Marks	50
Assignment / Quiz (s)	10+10
Teacher's Assessment	5

Assessment Components	CO1	CO2	CO3	CO4
Mid Sem Examination Marks	√	√	√	√
End Sem Examination Marks	√	√	√	√
Quiz I	√	√	√	
Quiz II	√	√	√	

Indirect Assessment –

1. Student Feedback on Faculty
2. Student Feedback on Course Outcome

Mapping between Objectives and Outcomes

Mapping of Course Outcomes onto Program Outcomes

Course Outcome #	Program Outcomes											
	a	b	c	d	e	f	g	h	i	j	k	l
1	3	3	3	3	1	1	1	2	1	1	1	1
2	3	3	3	3	1	1	1	2	1	1	1	1
3	1	3	3	3		1	1	1		1	1	1
4	2	2	2	2		2	2	2		1	1	2

Mapping Between COs and Course Delivery (CD) methods

		Course	Course Delivery
CD	Course Delivery methods	Outcome	Method

CD1	Lecture by use of boards/LCD projectors/OHP projectors	CO1, 2, 3, 4	CD1, CD2, CD3, CD8
CD2	Tutorials/Assignments	CO1, 2, 3, 4	CD1, CD2, CD3, CD8
CD3	Seminars		
CD4	Mini projects/Projects		
CD5	Laboratory experiments/teaching aids		
CD6	Industrial/guest lectures		
CD7	Industrial visits/in-plant training		
CD8	Self- learning such as use of NPTEL materials and internets		
CD9	Simulation		

